

Statistics & Causal Inference:

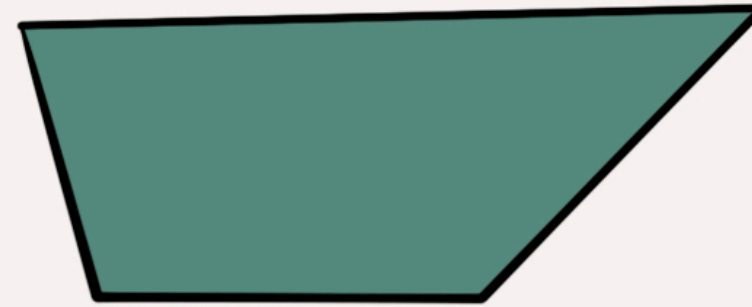
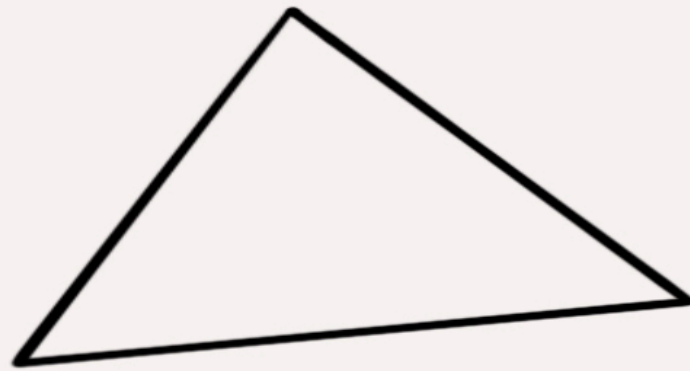
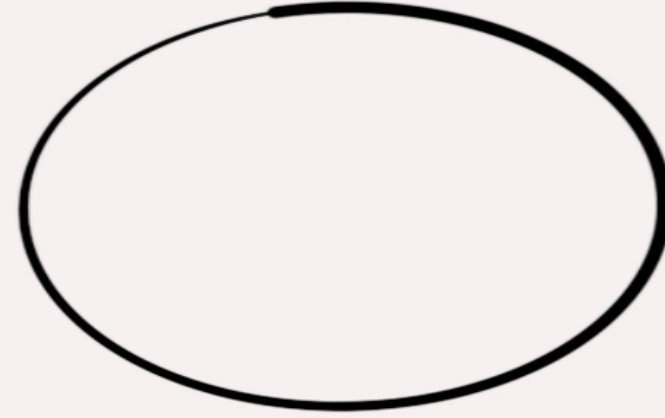
Neyman & Fisher



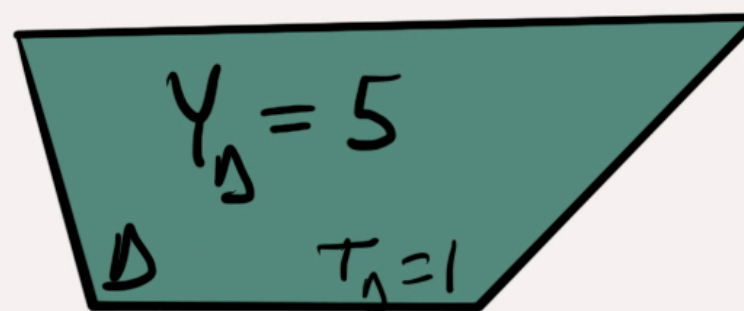
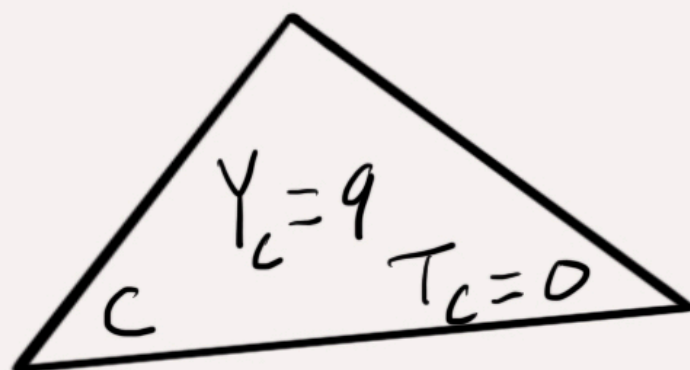
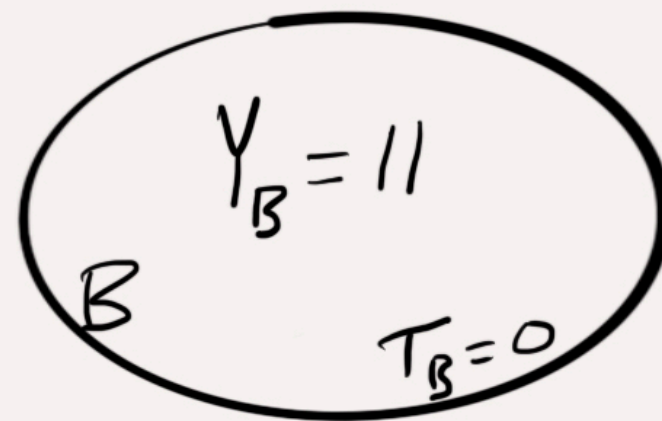
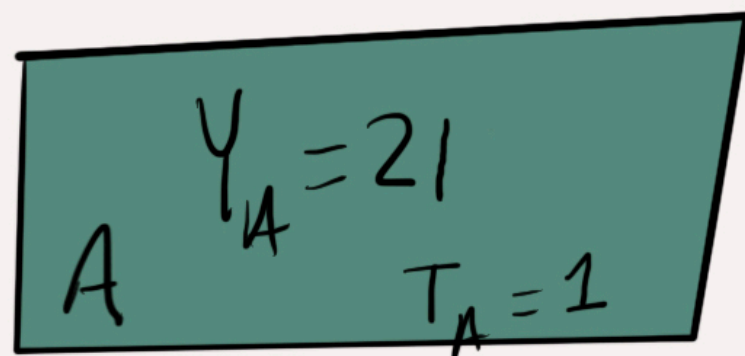
Jake Bowers
<http://jakebowers.org>

EGAP 2021
19 July 2021

What is the causal effect?

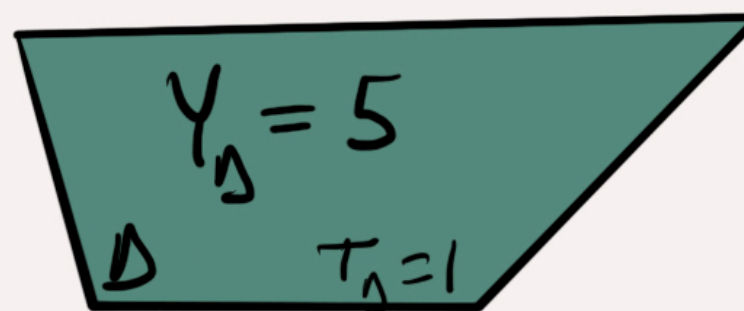
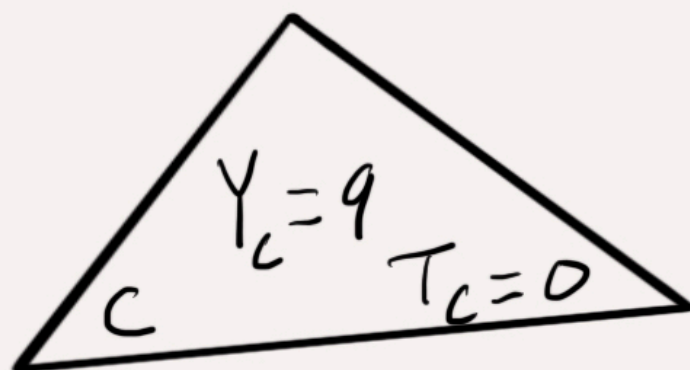
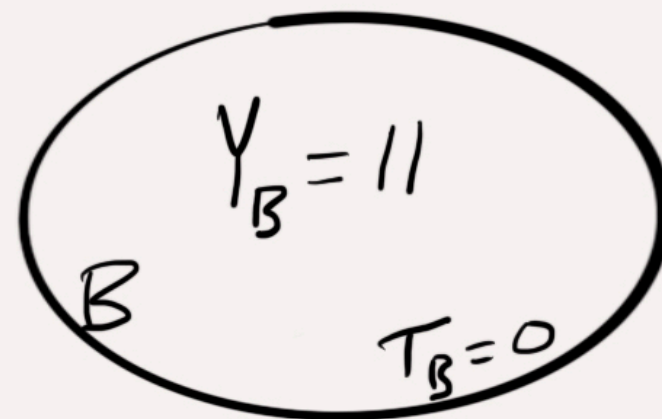
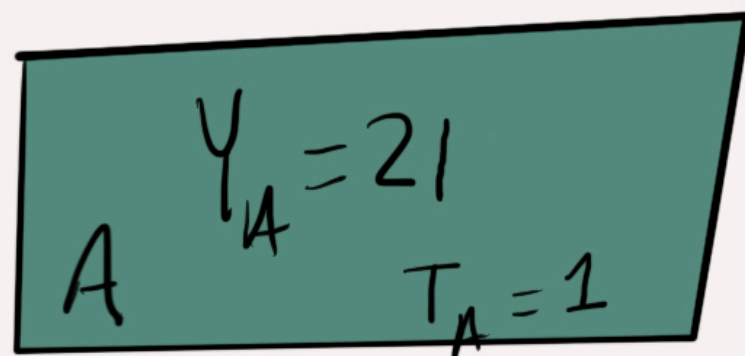


What is the causal effect?



i <u>Unit</u>	T <u>Treatment</u>	Y <u>Observed Outcome</u>
A	1	21
B	0	11
C	0	9
D	1	5

What is the causal effect?



i	T	Y	$Y(T=1)$	$Y(T=0)$	Effect (τ)
A	1	21	21	?	$21 - ? = \tau_A$
B	0	11	?	11	$? - 11 = \tau_B$
C	0	9	?	9	$? - 9 = \tau_C$
D	1	5	5	?	$5 - ? = \tau_D$

We don't know!



$$\begin{array}{c} Y_A = 21 \\ A \quad T_A = 1 \end{array}$$

$$\begin{array}{c} Y_B = 11 \\ B \quad T_B = 0 \end{array}$$



$$\begin{array}{c} Y_C = 9 \\ C \quad T_C = 0 \end{array}$$

$$\begin{array}{c} Y_D = 5 \\ D \quad T_D = 1 \end{array}$$

I don't know about the individual causal effects. BUT I can help you ESTIMATE the AVERAGE causal effect!

i	T	Y	$Y(T=1)$	$Y(T=0)$	Effect(τ)
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Define: $\bar{\tau} = (\tau_A + \tau_B + \tau_C + \tau_D) / 4$ (The avg. causal effect)

Notice: $\tau_A = Y_A(1) - Y_A(0)$, $\tau_B = Y_B(1) - Y_B(0)$, ...

$$\begin{aligned} \bar{\tau} &= ((Y_A(1) - Y_A(0)) + \dots + (Y_D(1) - Y_D(0))) / 4 \\ &= (Y_A(1) + \dots + Y_D(1)) / 4 - (Y_A(0) + \dots + Y_D(0)) / 4 \end{aligned}$$

$$\begin{array}{c} \text{A} \\ Y_A = 21 \\ T_A = 1 \end{array}$$

$$\begin{array}{c} \text{B} \\ Y_B = 11 \\ T_B = 0 \end{array}$$



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Define: $\bar{\tau} = (\tau_A + \tau_B + \tau_C + \tau_D) / 4$ (The avg. causal effect)
 an Unobserved Estimand $\left\{ \begin{array}{l} \bar{\tau} = (Y_A(1) + \dots + Y_D(1)) / 4 - (Y_A(0) + \dots + Y_D(0)) / 4 \\ \bar{\tau} = \bar{Y}(1) - \bar{Y}(0) \end{array} \right.$

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Define

an
Unobserved
Estimand $\bar{\tau} = (\tau_A + \tau_B + \tau_C + \tau_D) / 4$
 $\bar{\tau} = \bar{Y}(1) - \bar{Y}(0)$

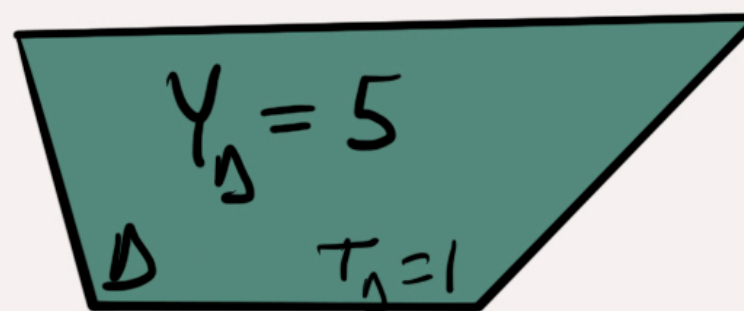
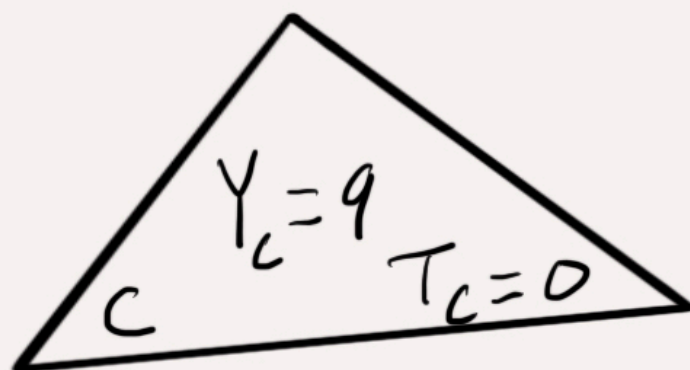
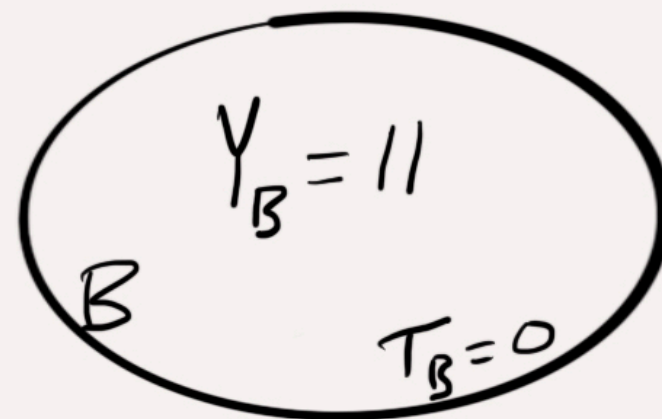
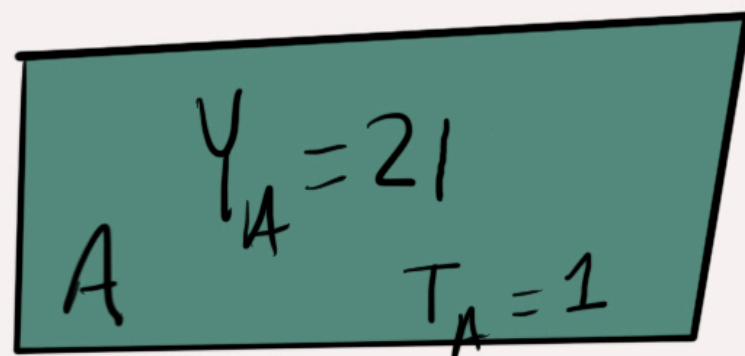
Choose

$$\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0) = \frac{(21+5)}{2} - \frac{(11+9)}{2} = 13 - 10 = 3$$

an
Estimator

(Show (Later) that $\hat{\tau}$ is a good estimator of $\bar{\tau}$)

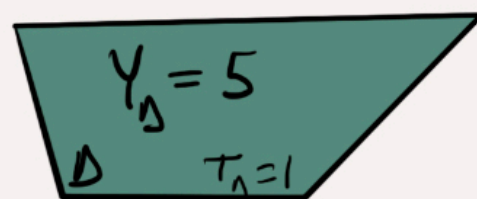
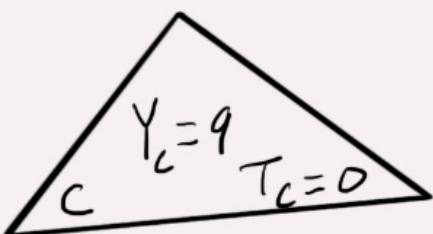
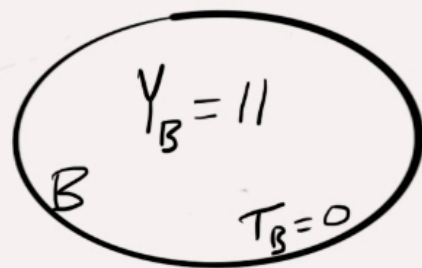
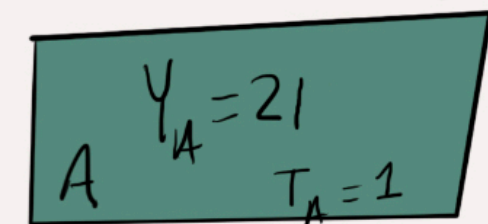
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C	0	9	?	9	$? - 9 = \tau_C$
D	1	5	5	?	$5 - ? = \tau_D$

We don't know!





I don't know about the individual causal effects. BUT I can help you TEST your ideas / HYPOTHESES about them.

An idea: "No effects"
 $H_0: Y_i(1) = Y_i(0)$

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$$\begin{matrix} Y_A = 21 \\ A \\ T_A = 1 \end{matrix}$$

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 $H_0: Y_i(1) = Y_i(0)$

What does this idea imply?

Notice: $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$

So H_0 implies: $Y_i = T_i \{Y_i(0)\} + (1 - T_i) Y_i(0)$
 $= Y_i(0)$

i	T	Y	$Y(T=1)$	$Y(T=0)$	Effect (τ)
A	1	21	21	21	$21 - ? = \tau_A$
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 $H_0: Y_i(1) = Y_i(0)$

$$H_0 \Rightarrow Y_i = Y_i(0)_{\text{observed}}$$

i	T	Y	$Y(T=1)$	$Y(T=0)$	Effect (τ)
A	1	21	21	21	$21 - ? = \tau_A$
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C	0	9	?	9	$? - 9 = \tau_C$
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So: $\frac{(21+5)}{2} - \frac{(11+9)}{2} = 3$ is compatible with H_0 even if not $\emptyset$.

But, A & D are only Treated by chance.

If A & B were treated, $H_0 \Rightarrow \frac{(21+11)}{2} - \frac{(9+5)}{2} = 9$

If C & D were treated, $H_0 \Rightarrow \frac{(9+5)}{2} - \frac{(21+11)}{2} = -9$

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① Idea/ H_0 : $Y_i(1) = Y_i(0)$

② H_0 implies $Y_i = Y_i(0)$

③ H_0 implies that a test statistic

summarizing $T \rightarrow Y$

$$t(T, Y) = \left(\frac{21+5}{2} \right) - \left(\frac{11+9}{2} \right) = 3$$

But also these other possible $t(T, Y)$ s :

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i	$\tilde{T}_1$	$\tilde{T}_2$	$\tilde{T}_3$	$\tilde{T}_4$	$\tilde{T}_5$	$\tilde{T}_6$
A	1	1	1	0	0	0
B	0	1	0	1	1	0
C	0	0	1	1	0	1
D	1	0	0	0	1	1
	3	9	7	-3	-7	-9

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① Idea/ H_0 : $Y_i(1) = Y_i(0)$

② H_0 implies $Y_i = Y_i(0)$

③ H_0 and Randomization imply 6 equally likely $\pm(T, Y)$ values.

i	T	Y	$Y(T=1)$	$Y(T=0)$	Effect (τ)
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D	1	5	5	5	$5 - ? = \tau_D$

In this experiment, "No effects" means

i	$\tilde{T}_1$	$\tilde{T}_2$	$\tilde{T}_3$	$\tilde{T}_4$	$\tilde{T}_5$	$\tilde{T}_6$
A	1	1	1	0	0	0
B	0	1	0	1	1	0
C	0	0	1	1	0	1
D	1	0	0	0	1	1

D 3 9 7 -3 -7 -9

$\pm(T, Y)$
no effects

A $Y_A = 21$
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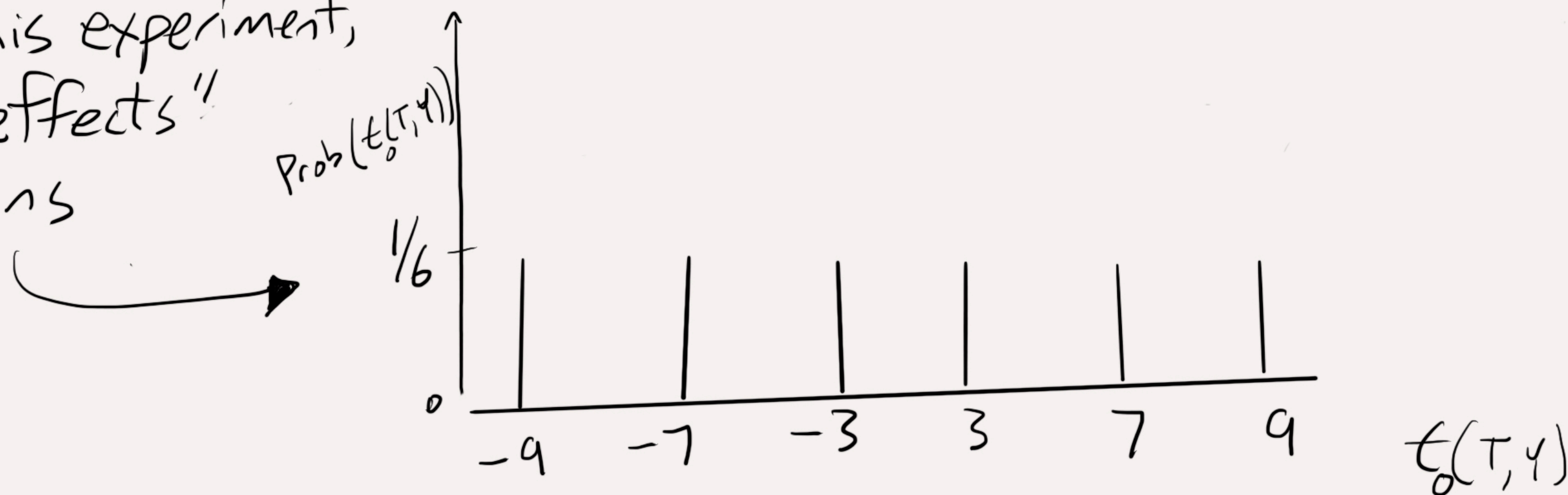
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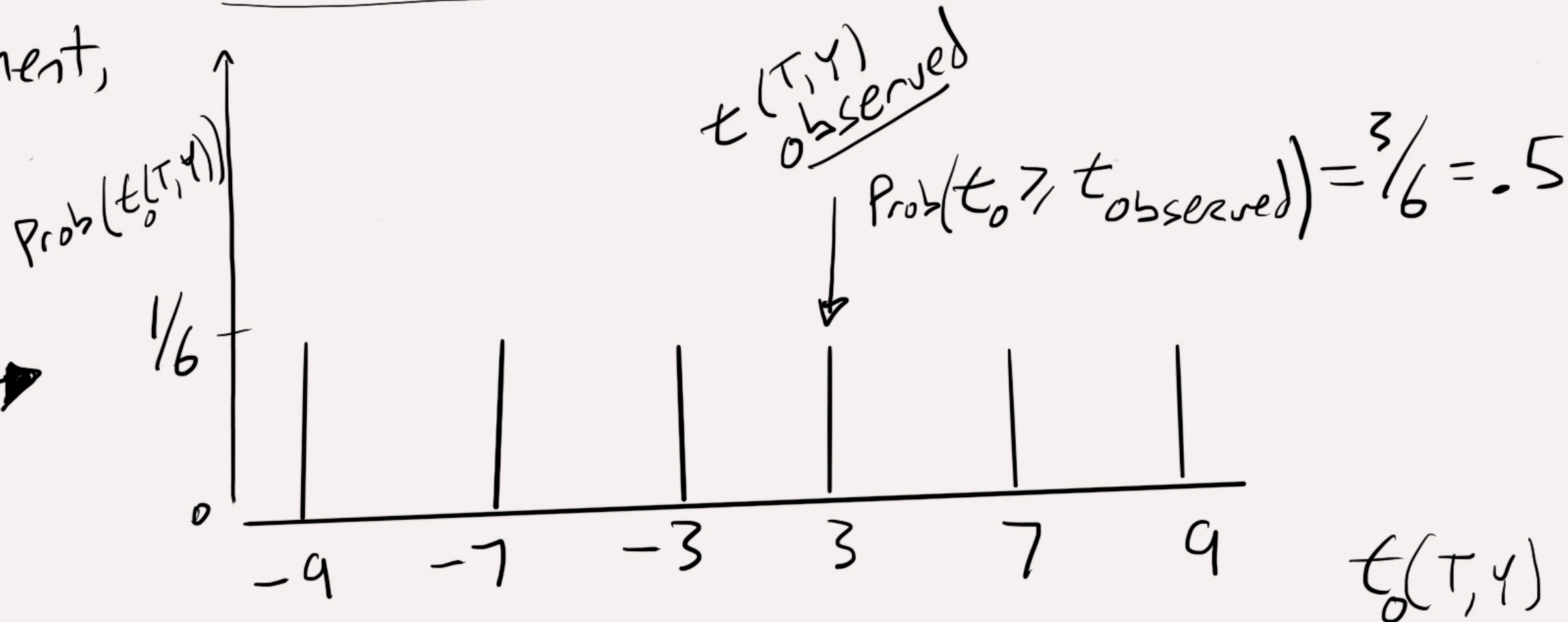
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I don't know about the individual causal effects. BUT I can help you TEST your ideas / HYPOTHESES about them.

① Idea/H₀: $Y_i(1) = Y_i(0)$

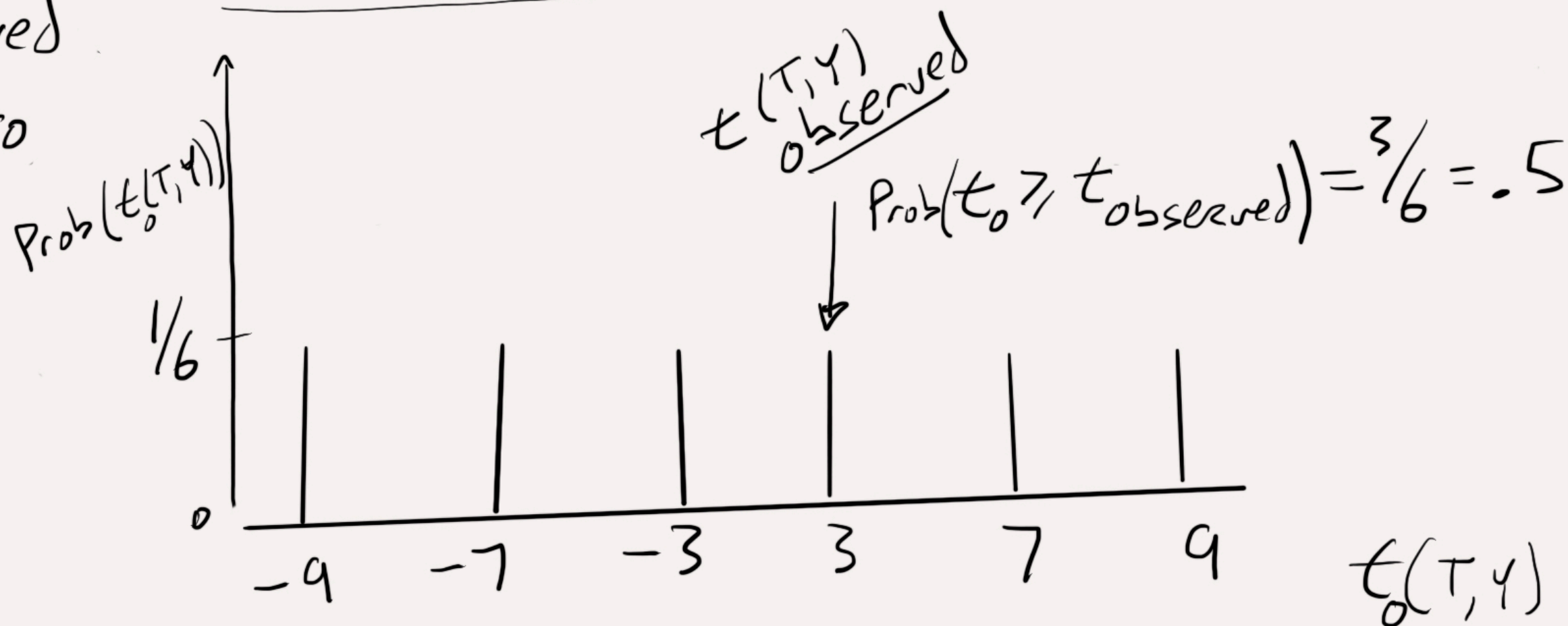
② H₀ implies $Y_i = Y_i(0)$

③ H₀ and Randomization imply 6 equally likely $t_0(T, Y)$ values.

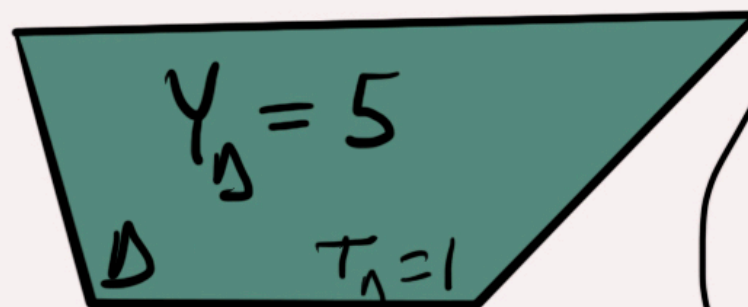
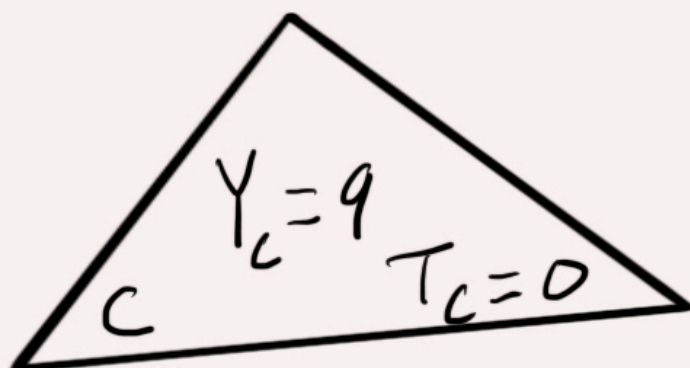
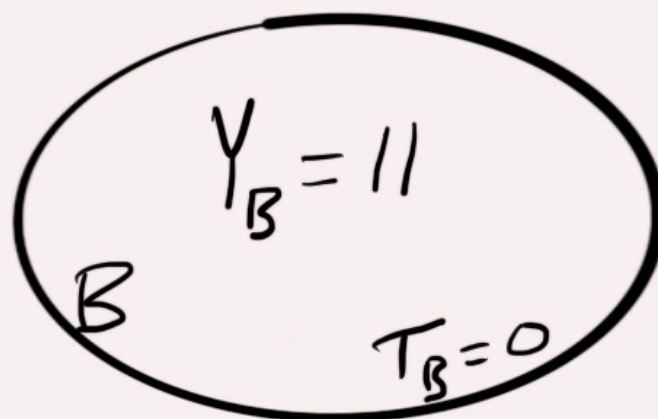
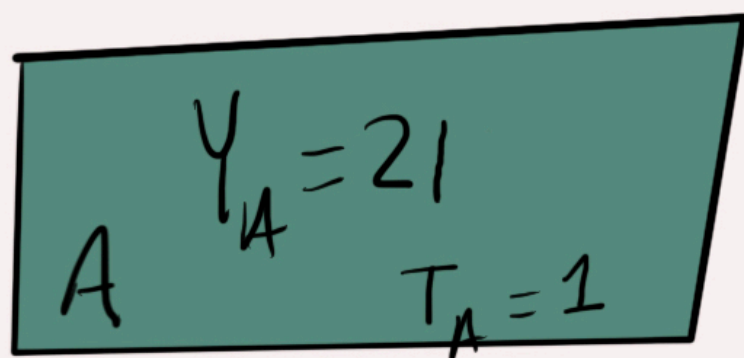
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④ Compare observed to implied by H₀ using a p-value

⑤ (Later) Show that this is a good test



What is the causal effect?



We don't know!
But...

You can estimate the average of the τ_i .

You can test ideas about the τ_i .

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